MS&E 228 (Winter 2026) — Lecture 19 Notes

Surrogates for Long-Term Treatment Effects

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March 20, 2026

Readings. *Applied Causal Inference Powered by ML and AI*, Chapter 15. Key papers: Athey, Chetty, Imbens & Kang (2019/2025); Prentice (1989); Battocchi, Dillon, Hei, Lewis, Oprescu & Syrgkanis (2021); Chen & Ritzwoller (2023); Chernozhukov, Newey, Singh & Syrgkanis (2022).

1 Introduction: The Challenge of Long-Term Outcomes

In many real-world applications of causal inference, the outcomes we truly care about unfold over long time horizons. Whether evaluating the efficacy of a new drug, the impact of a job training program, or the return on a marketing investment, the ultimate goal is to understand the long-term consequences of an intervention. However, waiting for these long-term outcomes to materialize is often impractical, costly, or simply too slow for decision-making. This chapter introduces the concept of **surrogate variables**—a powerful strategy for estimating long-term treatment effects using short-term data.

The central idea is to identify short-term signals, or surrogates, that act as **complete mediators** of the treatment's effect on the long-term outcome. By modeling the relationship between these surrogates and the final outcome, we can forecast the long-term effect long before it is directly observable. This approach has found wide application in diverse fields, from accelerating drug approvals in healthcare to optimizing online platforms in the tech industry.

The goals for this chapter are to:

1. Motivate the need for surrogate analysis with real-world examples from business, public policy, healthcare, and technology.
2. Formalize the identification of long-term effects using surrogate variables under both experimental and observational settings.
3. Introduce the **surrogate index**, a forecasting model that predicts long-term outcomes from short-term signals.
4. Develop a **doubly robust estimator** for the long-term average treatment effect that is resilient to model misspecification.

5. Discuss the critical assumptions underpinning surrogate analysis, including the potential for confounded surrogates.

1.1 Motivating Example: Estimating Long-Term Returns on Investment

A common problem faced by businesses is evaluating the long-term return on investment (ROI) of various customer-focused programs. For instance, a company like Microsoft might deploy new discount programs, assign dedicated sales specialists, or offer cloud computing credits to its customers. Each of these programs represents an investment with some cost, and the company wants to understand which ones are most successful in driving long-term customer value.

Figure 1: The core challenge: estimating long-term effects from short-term data.

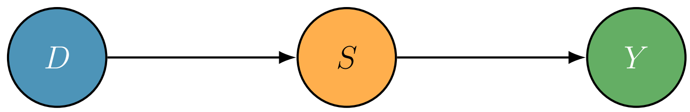
The primary outcome of interest is often a long-term metric, such as two-year customer revenue. However, a business cannot afford to wait two years to evaluate a program. It needs to make timely decisions about whether to continue, expand, or terminate these investments. This raises a critical question: **Can we construct reliable estimates of the long-term effects of these programs using only short-term data, for example, after just six months?**

Consider the specific example of an Azure credit program. Microsoft offers cloud computing credits to encourage customers to adopt the Azure platform. The hypothesis is that these credits will lead to long-term cloud adoption and increased spending. The outcome of interest is two-year revenue. We do not want to wait two years to determine whether the program should continue. The question is whether there is a way to construct estimates of the effect of the program on two-year revenue after, say, just six months.

This is precisely the problem that surrogate analysis is designed to solve. The key is to leverage short-term signals that are indicative of long-term success. In this example, the short-term signals might be the monthly purchase patterns of customers over the first six months. If the Azure credits are not utilized within the first six months—that is, if customers do not onboard onto the cloud platform during this period—then it is reasonable to assume that the credits will not affect long-term revenue. Conversely, if customers do onboard, their six-month purchasing behavior will be predictive of their long-term spending trajectory.

1.2 What Are Surrogates?

Surrogates are short-term signals S that are indicative of a customer's (or subject's) long-term reward Y. More precisely, surrogates are variables that act as **complete mediators** of the treatment's effect on the long-term outcome. This means that the treatment D affects the long-term outcome Y *if and only if* it affects these short-term signals S.



The typical approach is to take observations of the outcome variable itself over the short term. For instance, if the long-term outcome is two-year revenue, the surrogates might be the monthly revenue figures over the first six months. The willingness to use these as surrogates amounts to the assumption that the treatment will only affect the long-term revenue if and only if it affects those first six months of purchase volumes.

Returning to the Azure credits example: if the credits are offered and the customer does not spend them within the first six months, there is no mechanism by which those credits will affect long-term revenue. The effect of the credits on the customer's long-term journey must pass through the six-month purchasing behavior. If the customer onboarded onto the cloud in the first six months, that will have some projected or forecasted long-term revenue after two years.

The overall goal of this chapter is to understand how we can use those short-term signals to estimate the effect of the treatment on the long-term outcome.

2 Real-World Applications of Surrogates

The problem of estimating long-term effects is not unique to business operations; it is a pervasive challenge across many domains. Surrogates have become a standard tool in public policy, technology, and healthcare.

2.1 Application: Job Training Programs (Public Policy)

A seminal paper by Athey, Chetty, Imbens, and Kang (2019, published in the *Review of Economic Studies* in 2025) [1] demonstrated the power of surrogates in the context of public policy. The study re-analyzed the Greater Avenues for Independence (GAIN) job training program in California from the 1980s, which was designed to help welfare recipients find employment.

- **Treatment**: Job assistance, including search help, education, and training.
- **Long-term outcome**: 9-year post-program earnings.
- **Surrogates**: Employment rates and earnings over the first six quarters (1.5 years).

The original evaluation required waiting nearly a decade to measure the program's full impact. The authors showed that by using the first 1.5 years of employment data as surrogates, they could accurately predict the long-term earnings effects. Their surrogate-based estimates closely matched the true effects that were only observable after nine years. Specifically, the study plotted the estimated treatment effect as a function of the number of quarters of surrogate data used. After

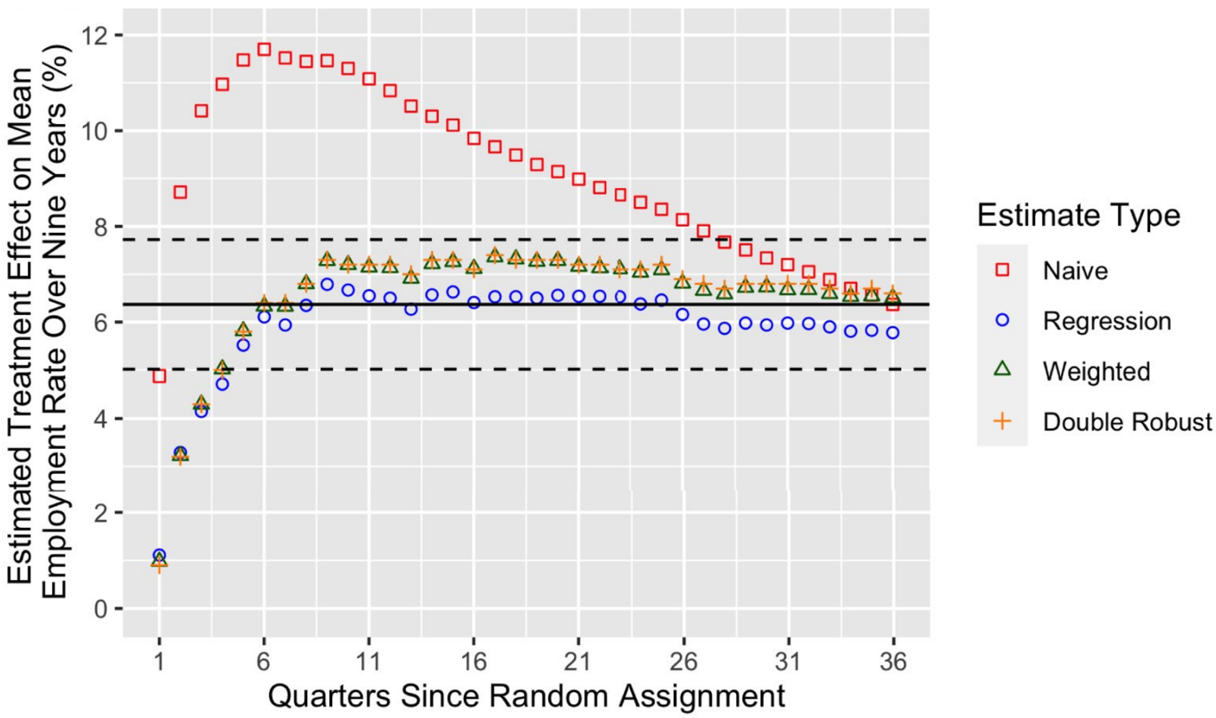


Figure 2: Empirical results from Athey, Chetty, Imbens and Kang, 2019 [1]. Each point is the estimated effect by various methods, as a function of the length of the surrogate signals in terms of quarters since random assignment. Naive: simply the effect on short term employment. Regression, Weighted, Double Robust: various methods that all should be recovering the correct effect in the large sample limit, but have different estimation properties in finite samples. Analogues of g-estimator, IPW estimator and DR estimator, we've seen in the past. We will elaborate on these estimators later in the lecture.

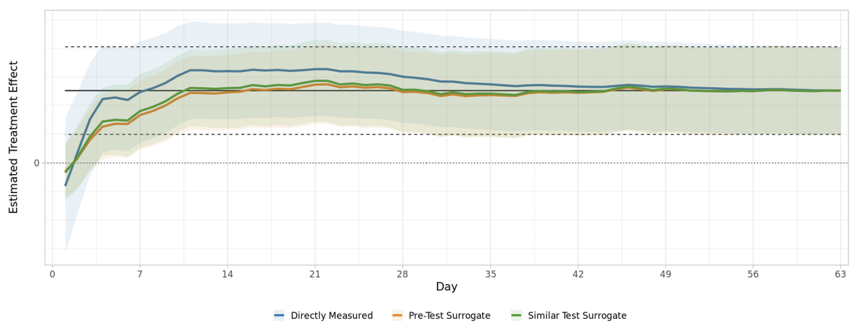
approximately six quarters, the surrogate-based estimate converged to the true long-term effect (the black line in Figure 2). This finding was a powerful demonstration that surrogate analysis can dramatically reduce the time required for policy evaluation—from nearly a decade to just about one and a half years—enabling policymakers to learn and iterate much more quickly.

2.2 Application: A/B Testing at Netflix (Tech)

Another domain where surrogates are used daily is in the technology industry. It is well known that companies like Netflix [9], Amazon [10], and virtually every major company with an experimentation platform use surrogates to understand the long-term effects of their experiments. When an A/B test is run, the goal is to understand the effect on long-term metrics such as two-year revenue, retention, or engagement. Short-term surrogates, like the response of the user in the next month, are used to forecast the long-term effect.

A paper by Zhang, Zhao, Le, and Kallus (2023) [9] analyzed 200 A/B tests at Netflix that evaluated changes to the platform's personalization algorithms.

- **Treatment**: Changes to content recommendation algorithms or user interface.
- **Long-term outcome**: Day 63 member retention and engagement.
- **Surrogates**: Short-term (14-day) observations of user behavior, such as streaming hours or click-through rates.



The study found that decisions made using a 14-day surrogate index were approximately 95% consistent with the decisions that would have been made by waiting for the full 63-day outcome. The same type of convergence plot was observed: as a function of the number of days of surrogate data used, the estimated effect converged to the true long-term effect. After about 14–15 days, the surrogate-based estimate matched the 63-day outcome very well. This allows Netflix to accelerate its experimentation cycle, quickly identifying which product changes are likely to have a positive long-term impact on user engagement and retention.

2.3 Application: Accelerated Drug Approval (Healthcare)

The concept of surrogates has its historical roots in healthcare. The first seminal paper that introduced the idea of surrogates, by Ross Prentice in 1989, was motivated by the challenge of evaluating drugs for chronic diseases. The U.S. Food and Drug Administration (FDA) does not want to wait 20 years to observe long-term survival outcomes for chronic diseases. Instead, when evaluating a drug, the FDA identifies short-term biomarkers—such as immune responses, CD4 cell counts, or tumor shrinkage—that are indicative of long-term survival. These short-term biomarkers are then used to measure the success of the drugs and decide whether or not to approve them.

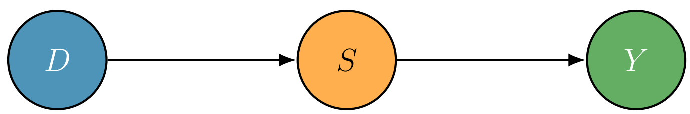
- **Treatment**: Novel drugs and therapies for diseases like HIV, cancer, or diabetes.
- **Long-term outcome**: Overall survival, or the occurrence of major adverse events.

- **Surrogates**: Biological or physiological markers (biomarkers) that are thought to predict clinical benefit. Examples include CD4 cell counts for HIV, tumor shrinkage for cancer, or HbA1c levels for diabetes.

The FDA's Accelerated Approval Pathway explicitly uses the concept of surrogates. This pathway allows life-saving drugs to become available to patients years earlier than they would under the traditional approval process. For example, many early antiretroviral therapies for HIV were approved based on their ability to increase CD4 counts [11], long before data on their long-term effect on survival was available.

3 Identification of Long-Term Effects with Surrogates

Now that we have established the broad applicability of surrogate analysis, we turn to the formal identification of long-term treatment effects. The core assumption is that the short-term surrogates, S, act as a **complete mediator** of the treatment's effect on the long-term outcome, Y. This is captured by the following causal graph:



This DAG implies the conditional independence relationship $Y \perp\!\!\!\perp D \mid S$. In words, once we know the value of the short-term surrogates, the treatment itself provides no additional information about the long-term outcome. The treatment affects the long-term outcome *if and only if* it affects the short-term surrogates.

3.1 The Identification Strategy: Experimental Setting

Consider the simplest case: a randomized controlled trial (A/B test) where the treatment D is exogenously assigned. This is the Netflix setting, where an A/B test is run and the goal is to estimate the effect on the two-month outcome, with S representing the 14-day behavior of the user. Our goal is to estimate the average treatment effect, ATE $= \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)]$. The key insight, which we also encountered in the previous chapter on dynamic treatment effects, is that the effect on the long-term outcome is equal to the effect on the *forecasted* long-term outcome, where the forecast is based on the surrogates. If there is a complete mediator, the effect of the treatment on the outcome is the effect of the treatment on the forecasted outcome from that complete mediator.

Let us derive this formally for the average potential outcome under treatment level d, $\mathbb{E}[Y(d)]$. Since the treatment is randomized, we have $\mathbb{E}[Y(d)] = \mathbb{E}[Y \mid D = d]$ by exogeneity. We can then apply the law of total expectation (tower rule), conditioning on the surrogates:

$$\begin{aligned} \mathbb{E}[Y(d)] &= \mathbb{E}[Y \mid D = d] && \text{(Exogeneity of Treatment)} \\ &= \mathbb{E}\big[\mathbb{E}[Y \mid D = d, S] \,\big|\, D = d\big] && \text{(Tower Rule)} \end{aligned}$$

Now, we invoke the key surrogacy assumption, $Y \perp\!\!\!\perp D \mid S$. Looking at the DAG, when we condition on the surrogates S, the treatment D is d-separated from the outcome Y. This allows us to drop the conditioning on D inside the inner expectation:

$$\mathbb{E}[Y(d)] = \mathbb{E}\Big[\underbrace{\mathbb{E}[Y \mid S]}_{\substack{\text{Forecast of } Y \\ \text{from } S}} \Big| D = d\Big] \qquad \text{(Surrogacy: } Y \perp\!\!\!\perp D \mid S\text{)}$$

Let us define the **surrogate index** as the function

$$h(S) = \mathbb{E}[Y \mid S] \qquad \text{(surrogate index)}$$

This function represents the best possible forecast of the long-term outcome Y using only the information contained in the short-term surrogates S. With this definition, the identification result becomes:

Surrogate Identification Formula (Experimental Case)

$$\mathbb{E}[Y(d)] = \mathbb{E}[h(S) \mid D = d]$$

where $h(S) = \mathbb{E}[Y \mid S]$ is the surrogate index.

This result is powerful. It states that to estimate the long-term effect, we do not need to observe the long-term outcome Y in our main experimental dataset. Instead, we can:

1. Build a forecasting model $h(S)$ that, given the surrogates, predicts the long-term reward.
2. In the short-term experimental data, use the learned model $h(S)$ to forecast long-term rewards for each treated and control individual based on their short-term surrogate values.
3. Estimate the treatment effect on these imputed outcomes. For a randomized experiment, this is simply the difference in the average imputed outcomes between the treatment and control groups:
$$\widehat{\text{ATE}} = \frac{1}{n_1} \sum_{i:D_i=1} h(S_i) - \frac{1}{n_0} \sum_{i:D_i=0} h(S_i)$$

3.2 The Two-Dataset Problem and the Surrogate Index

A natural question arises: how do we build the forecasting model $h(S) = \mathbb{E}[Y \mid S]$? We cannot build it from the experimental data, because in the experimental data we do not observe the long-term outcome Y—we are still at the 14-day mark. We need to use some *other* data, separate from the experimental data, where both the short-term surrogates S and the long-term outcome Y are observed.

The practical implementation of this strategy therefore requires two distinct datasets:

- **Short-term (experimental) data** (s): Contains the treatment D and the surrogates S, but *not* the long-term outcome Y. This is the data from the current A/B test or observational study.
- **Long-term (historical) data** (ℓ): Contains the surrogates S and the long-term outcome Y. The treatment D may not even exist in this dataset. This data is used to train the surrogate index model $h(S) = \mathbb{E}_\ell[Y \mid S]$.

Long-term data (ℓ) **Short-term data (s)**

Where does the long-term historical data come from? In the Netflix example, two approaches were used. First, the same users were observed but from the *previous* two months: a forecasting model was built using the same users' data from the last two months, predicting two-month outcomes from 14-day behavior. Second, data from other similar experiments that happened in the past was used. In the past, similar features may have been tested, and for those historical experiments, the long-term two-month rewards are already observed. By concatenating all such similar experiments and examining 14-day behavior alongside two-month rewards, a regression model was built.

3.2.1 The Invariance Assumption

This two-dataset structure introduces a critical, untestable assumption: **invariance**. We must assume that the relationship between the surrogates and the long-term outcome is the same in both the historical and the current experimental population. Formally:

Invariance Assumption

$$\mathbb{E}_s[Y \mid S] = \mathbb{E}_\ell[Y \mid S]$$

The conditional expectation of the long-term outcome given the surrogates is the same in the short-term (experimental) data distribution and the long-term (historical) data distribution.

It is important to note that we do *not* need the populations to be the same in terms of the distribution of the surrogates. What matters is that the *relationship* between short-term and long-term outcomes is invariant. How those users behaved in the last two months—the relationship between their 14-day behavior and the two-month spend—should be the same in the last two months as in the next two months.

When Might the Invariance Assumption Fail?

The invariance assumption can be violated if there are systematic changes in the data-generating process over time. For example, if there is a market trend of consistently growing revenue, the relationship between 6-month purchasing behavior (S) and 2-year revenue (Y) from two years ago might be different from that relationship today. If you look at historical data, they are at some point on that growing curve; in the next two months, there is another point on the growing curve. So the conditional expectation will actually change because there is some time trend.

A practical heuristic to mitigate this is to first **detrend** the data: calculate some market trends and subtract those market trends. After that, the invariance holds, at least approximately. So one can subtract those macro trends before learning the surrogate index model.

How to Validate the Invariance and Surrogacy Assumptions

While there are no formal statistical tests for the invariance or the surrogacy assumptions, a practical validation strategy is to use historical experiments. One can take a pool of historical experiments where the long-term outcome is observed, hold one experiment out of sample, and ask: if we use the rest of the experiments to build a surrogate model, can we predict the long-term effect of the held-out experiment as accurately as if we had used the long-term outcome directly?

This is typically done with a growing window of surrogates. The validation plot shows the estimated effect as a function of the number of time periods of surrogate data used. After some window of surrogates, the surrogate-based estimate converges to the true long-term effect. This is also how one chooses the appropriate window of surrogates in practice.

Once we believe in the invariance assumption, the procedure is:

1. Go to the historical data and build an ML predictive model that predicts the long-term reward

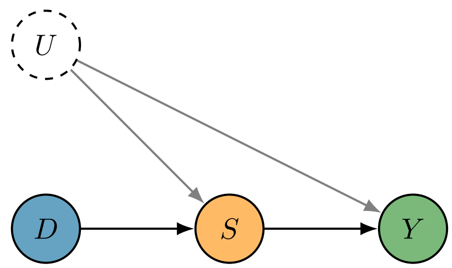
from the short-term behavior. This model is called the **surrogate index**.

2. Go to the short-term data and use the surrogate index model to forecast imputed long-term rewards for each individual.

3. Estimate the effect of the treatment on those forecasted long-term rewards.

3.3 Caution: Confounded Surrogates

The identification strategy relies critically on the causal structure $D \to S \to Y$. The argument breaks down if there is an unobserved confounder U that affects both the surrogates S and the long-term outcome Y, as shown below.



In this scenario, the key conditional independence $Y \perp\!\!\!\perp D \mid S$ **does not hold**. The key property that was used in the chain of equations was precisely this independence. Does it hold in this graph? No. Using the d-separation criteria, conditioning on the surrogate S—which is a collider on the path $D \to S \leftarrow U \to Y$—opens a spurious association between D and Y. Therefore, the identification formula $\mathbb{E}[Y(d)] = \mathbb{E}[\mathbb{E}[Y \mid S] \mid D = d]$ is no longer valid. To block this confounding path, one would also need to condition on the confounder U of the surrogates, so that $Y \perp\!\!\!\perp D \mid S, U$ holds.

The Danger of Spurious Correlation in Surrogates

Consider a scenario in a tech company where highly engaged users tend to click more (S) and also tend to have higher downstream outcomes (Y). If one naively uses short-term clicks to predict long-term reward, the effect of the treatment will be exaggerated. A treatment might drive certain clicks, and those short-term clicks will be correlated with high long-term reward—but not because the clicks themselves drive the long-term reward. Rather, it is because the people who are highly engaged click more.

Ideally, what one really wants to learn in a surrogate method is the **causal edge** from S to Y only, not the spurious association driven by the confounder. If the surrogates are confounded, the surrogate index model $h(S) = \mathbb{E}[Y \mid S]$ will pick up both the causal and the spurious association, leading to biased estimates.

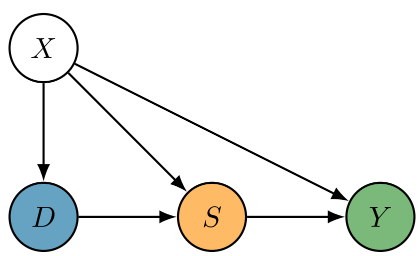
3.3.1 Dealing with Confounded Surrogates

If the surrogates are confounded, an alternative approach is to try to isolate only the causal relationship between the surrogates and the outcome, and remove the spurious association. One needs to learn the *causal effect* of the surrogates on the long-term reward—not just the predictive model, but the causal model—and then use that causal association when measuring the long-term effect.

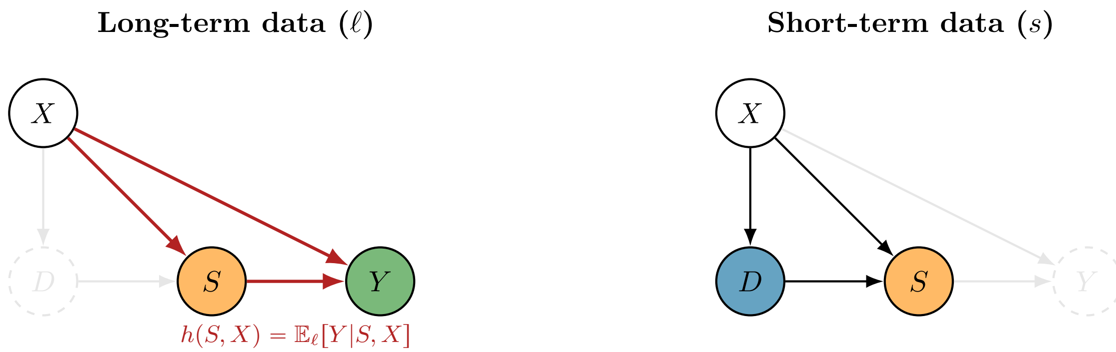
What many people have done in experimentation platforms like Netflix is to use a collection of other experiments as **instruments**. These are experiments that drive the surrogates. Because these experiments are exogenous (randomly assigned), and the surrogates are a complete mediator (meaning these experiments do not have a direct effect on the outcome), the exclusion restriction of the instrument is satisfied. Therefore, instrumental variable regression can be used to estimate the causal effect of S on Y. The learned causal effect model can then be used as the surrogate index model when estimating the effect of a new experiment (see e.g. Bibaut, Kallus, Ejdemyr & Zhao, 2023 [4]). There is also a very nice recent set of papers (e.g., Imbens, Kallus, Mao & Wang, 2025 [7]) that describes other techniques for dealing with confounded surrogates. Interestingly, these techniques resemble proximal causal inference, which was covered in a previous chapter.

3.4 Identification in Observational Settings

For the remainder of this chapter, we focus on the situation where we observe all of the confounders—both of the surrogates and of the treatment. The surrogate framework extends naturally to observational studies where the treatment D is not randomized but is confounded by a set of observed covariates X. This applies not just for A/B tests, like in the Netflix scenario, but also for observational studies, like in the job training example or in the Microsoft example. The causal graph becomes:



Here, we assume conditional exogeneity of the treatment given the covariates, $Y(d) \perp\!\!\!\perp D \mid X$, and also that the surrogacy assumption holds conditional on X, i.e., $Y \perp\!\!\!\perp D \mid S, X$. When building the surrogate index model (the forecasting model), we now also need to include the confounders of either the surrogates or the treatment. So we include those confounders and build a bigger surrogate index model that includes both the short-term behavior S and also all of the other pre-treatment drivers X:



The identification proceeds by applying the g-formula, but with the outcome replaced by the surrogate index. Let us derive this step by step. We want to estimate the average potential outcome under treatment level d:

$$\tau(d) := \mathbb{E}_s[Y(d)] = \mathbb{E}_s\big[\mathbb{E}_s[Y \mid D = d, X]\big] \tag{g-formula, $Y(d) \perp\!\!\!\perp D \mid X$}$$

Now, we use the tower rule to introduce the surrogates into the inner expectation:

$$\mathbb{E}_s[Y \mid D = d, X] = \mathbb{E}_s\big[\mathbb{E}_s[Y \mid D, S, X] \,\big|\, D = d, X\big] \tag{Tower Rule}$$

By the surrogacy assumption $Y \perp\!\!\!\perp D \mid S, X$, we can drop the conditioning on D in the innermost expectation:

$$\mathbb{E}_s[Y \mid D = d, X] = \mathbb{E}_s\big[\mathbb{E}_s[Y \mid S, X] \,\big|\, D = d, X\big] \tag{Surrogacy: $Y \perp\!\!\!\perp D \mid S, X$}$$

Finally, by the invariance assumption $\mathbb{E}_s[Y \mid S, X] = \mathbb{E}_\ell[Y \mid S, X] = h(S, X)$, we obtain:

Surrogate Identification Formula (Observational Case)

$$\tau(d) = \mathbb{E}_s\big[\mathbb{E}_s[h(S, X) \mid D = d, X]\big]$$

where $h(S, X) = \mathbb{E}_\ell[Y \mid S, X]$ is the surrogate index trained on the long-term data.

This is a g-formula where the outcome regression is not predicting the long-term outcome directly, but rather predicting the forecasts of the surrogate index model. The surrogate index model itself is trained on the separate long-term dataset to predict the long-term outcome from the surrogates and the covariates.

3.5 The Nested Regression Estimand

The identification formula has a **nested regression** structure. It involves two levels of regression:

$$\tau(d) = \mathbb{E}_s[g(d, X)]$$
$$g(D, X) = \mathbb{E}_s[h(S, X) \mid D, X]$$
$$h(S, X) = \mathbb{E}_\ell[Y \mid S, X]$$

- **Outer Regression (g)**: $g(D, X) = \mathbb{E}_s[h(S, X) \mid D, X]$. This is learned on the short-term data, regressing the imputed outcomes $h(S, X)$ on the treatment D and covariates X.
- **Inner Regression (h)**: $h(S, X) = \mathbb{E}_\ell[Y \mid S, X]$. This is the surrogate index, learned on the long-term data, regressing the long-term outcome Y on the surrogates S and covariates X.
- The target parameter is averaged over the short-term data distribution.
- The g regression function is trained on the outputs of the h regression function.
- The surrogate index model h is trained on the long-term data distribution.

This structure is qualitatively similar to the Q-function structure we encountered in the previous chapter on dynamic treatment effects, where we had Q_1 and Q_2. Here, we have g and h playing analogous roles. There is also an extra difficulty: the forecasting model h is not even trained on the same dataset as the g model—it is trained on different historical data.

4 Doubly Robust Estimation of Long-Term Effects

A simple plug-in estimator—estimating the nuisance functions $\hat{h}$ and $\hat{g}$ and plugging them into the formula—can be sensitive to errors in these regressions. Because the target quantity is some formula that depends on a regression, we will add debiasing or correction terms, like the doubly robust term. This is the exact same style of robust formulas that we have already seen, including in the dynamic treatment effects chapter.

4.1 The Debiased Formula: First Correction

We start with the nested estimand for a specific treatment level d:

$$\tau(d) = \mathbb{E}_s[g(d, X)] \quad \text{where} \quad g(D, X) = \mathbb{E}_s[h(S, X)|D, X] \quad \text{and} \quad h(S, X) = \mathbb{E}_\ell[Y|S, X]$$

The first step is to correct for the estimation of $g(D, X)$. This is a standard debiasing step, adding an inverse propensity weighted (IPW) term for the residual of the g regression:

$$\tau(d) = \mathbb{E}_s\left[g(d,X) + \underbrace{\frac{\mathbf{1}\{D=d\}}{\Pr_s(D=d \mid X)}}_{\text{Inverse Propensity Weights}} \cdot \underbrace{(h(S,X) - g(D,X))}_{g\text{ regression residual}} \right]$$

This is the residual of the g regression, multiplied by the inverse propensity weights. However, this formula is not yet fully debiased because it still depends on the function $h(S, X)$, which is itself an estimate from a separate regression. By adding the debiasing term, we have introduced yet another regression function, and so we need to further debias this formula for that regression function.

4.2 Recursive Debiasing: Second Correction

We need to add a second correction term that uses the residuals of the surrogate index model and multiplies them by some weighting function. The complete, doubly robust formula for $\tau(d)$ is:

$$\tau(d) = \mathbb{E}_s\left[g(d,X) + \frac{\mathbf{1}\{D=d\}}{\Pr_s(D=d|X)}\big(h(S,X) - g(D,X)\big) \right] + \mathbb{E}_\ell\left[w_d(S,X) \cdot \underbrace{(Y - h(S,X))}_{h\text{ regression residual}} \right]$$

A key observation is that we only observe long-term outcomes Y in the historical data (ℓ). Therefore, the second correction term *must* be an expectation over the historical data. We cannot have a correction term that takes an expectation over the short-term data and uses the long-term outcome labels, because we do not observe those labels in the short-term data.

The weighting function $w_d(S, X)$ is of the form:

$$w_d(S,X) = s(S,X) \cdot \mathbb{E}_s\left[\frac{\mathbf{1}\{D=d\}}{\Pr_s(D=d \mid X)} \,\middle|\, S,X \right] = s(S,X) \cdot \frac{\Pr_s(D=d \mid S,X)}{\Pr_s(D=d \mid X)}$$

This weight has two components:

Surrogate Score $s(S, X)$. This is a density ratio between the short-term and long-term data distributions:

$$s(S,X) = \frac{p(S,X \mid E=s)}{p(S,X \mid E=\ell)} = \frac{\Pr(E=s \mid S,X)\ \Pr(E=\ell)}{\Pr(E=\ell \mid S,X)\ \Pr(E=s)}$$

where E is a binary indicator for the dataset identity ($E = s$ for short-term, $E = \ell$ for long-term). This ratio accounts for the potential **covariate shift** between the short-term and long-term datasets. Maybe the historical data have a different distribution of users than the short-term data. This ratio captures how similar these distributions are, or how probable a user with a given (S, X) profile was in the short-term versus the long-term data.

In practice, this density ratio can be estimated by training a classifier to distinguish between samples from the two datasets. We create a synthetic dataset where the label is the dataset identity (E), and we run a classifier that estimates the probability of being a sample from the short-term or the long-term data. This is a standard trick for reducing density ratio estimation to a classification problem.

Projected IPW. The term $\frac{\Pr_s(D=d|S,X)}{\Pr_s(D=d|X)}$ is the result of projecting the original IPW term onto the variables (S, X) available in the long-term data. In the long-term data, we do not observe the treatment D. So we cannot simply take the inverse propensity and put it in the correction term, because the inverse propensity depends on the treatment. But by doing this projection—taking the conditional expectation of the IPW given (S, X)—we replace it with the probability of treatment given the surrogate profile. We look at the surrogate profile in the historical data and ask: for such a surrogate profile, how probable is it that we would have treated this user? This quantity is learned in the short-term data, because in the short-term data we do have the treatment and the surrogate behavior.

4.3 Doubly Robust Formula for the ATE

To compute the Average Treatment Effect (ATE), we take the difference ATE $= \tau(1) - \tau(0)$. This results in a simplified expression where the IPW terms for the g-regression residual combine:

Doubly Robust Formula for Surrogate ATE

$$\theta = \mathbb{E}_s\left[g(1,X) - g(0,X) + \frac{D - p(X)}{p(X)\,(1-p(X))}\big(h(S,X) - g(D,X)\big)\right]$$
$$+\,\mathbb{E}_\ell\left[s(S,X)\left(\frac{\pi(S,X)}{p(X)} - \frac{1-\pi(S,X)}{1-p(X)}\right)\cdot(Y - h(S,X))\right]$$

where:

- $p(X) = \Pr_s(D = 1 \mid X)$ is the propensity score
- $\pi(S,X) = \Pr_s(D = 1 \mid S, X)$ is the extended propensity score
- $s(S,X)$ is the surrogate score (density ratio)
- $g(D,X) = \mathbb{E}_s[h(S,X) \mid D, X]$ is the outer regression
- $h(S,X) = \mathbb{E}_\ell[Y \mid S, X]$ is the surrogate index

The simplification in the first line uses the fact that $\frac{\mathbf{1}\{D=1\}}{p(X)} - \frac{\mathbf{1}\{D=0\}}{1-p(X)} = \frac{D-p(X)}{p(X)(1-p(X))}$, which is the standard IPW simplification for the ATE.

This formula is doubly robust. The estimate for θ will be consistent if either:

1. The models for the outcome regressions, $g(D, X)$ and $h(S, X)$, are correctly specified.

2. The models for all the propensity scores ($p(X)$, $\pi(S, X)$) and the density ratio $s(S, X)$ are correctly specified.

This provides a significant degree of protection against model misspecification, which is crucial in high-dimensional settings where flexible machine learning models are used.

4.4 Implementation via Cross-Fitting

To avoid overfitting and ensure valid statistical inference, all nuisance functions are estimated using **cross-fitting**. The procedure, as implemented in the accompanying Jupyter notebook, is as follows. The key design choice is that the cross-fitting is **coupled** across the short-term and long-term datasets: the combined data is split into K folds, and for each fold, all nuisance models are trained on the out-of-fold data and predictions are generated for the held-out fold. This ensures that all nuisance estimates are out-of-sample.

4.4.1 Nuisance Models

The following nuisance models need to be estimated:

Model	**Definition**	**Trained on**
$h(S, X)$	$\mathbb{E}_\ell[Y \mid S, X]$	Long-term data (ℓ)
$g(D, X)$	$\mathbb{E}_s[h(S, X) \mid D, X]$	Short-term data (s)
$p(X)$	$\Pr_s(D = 1 \mid X)$	Short-term data (s)
$\pi(S, X)$	$\Pr_s(D = 1 \mid S, X)$	Short-term data (s)
$s(S, X)$	Density ratio $p_s(S, X)/p_\ell(S, X)$	Combined data

4.4.2 Pseudocode

The following pseudocode implements the full DR Surrogate ATE estimator with cross-fitting.

DR Surrogate ATE Estimator: Pseudocode — Nuisance Estimation

```
def dr_surrogate_ate(data_s, data_l, S_cols, X_cols, K=5,
                     reg_model, clf_model):
    SX, DX = S_cols + X_cols, ["D"] + X_cols
    n_s, n_l = len(data_s), len(data_l)
    # Initialize arrays for out-of-sample predictions
    h_hat_l, h_hat_s = np.zeros(n_l), np.zeros(n_s)
    g_hat, g1_hat, g0_hat = np.zeros(n_s), np.zeros(n_s), np.zeros(n_s)
    pi_hat = np.zeros(n_s)
    w1_hat, w0_hat = np.zeros(n_l), np.zeros(n_l)
    # Label datasets: E=1 for short-term, E=0 for long-term
    data_s["E"], data_l["E"] = 1, 0
    data_all = pd.concat([data_l, data_s])  # first n_l rows: long-term
    # Coupled cross-fitting over combined data
    for train, test in KFold(n_splits=K, shuffle=True).split(data_all):
        tr_l, te_l = train[train < n_l], test[test < n_l]
        tr_s, te_s = train[train >= n_l] - n_l, test[test >= n_l] - n_l
        # --- h = E_l[Y | S, X]: surrogate index ---
        h_m = clone(reg_model).fit(
            data_l.iloc[tr_l][SX], data_l.iloc[tr_l]["Y"])
        h_hat_l[te_l] = h_m.predict(data_l.iloc[te_l][SX])
        h_hat_s[te_s] = h_m.predict(data_s.iloc[te_s][SX])
        # --- g = E_s[h(S,X) | D, X]: outer regression ---
        g_m = clone(reg_model).fit(
            data_s.iloc[tr_s][DX],
            h_m.predict(data_s.iloc[tr_s][SX]))
        g_hat[te_s] = g_m.predict(data_s.iloc[te_s][DX])
        X_te = data_s.iloc[te_s][X_cols]
        te_1 = X_te.copy(); te_1.insert(0, "D", 1)
        te_0 = X_te.copy(); te_0.insert(0, "D", 0)
        g1_hat[te_s] = g_m.predict(te_1)
        g0_hat[te_s] = g_m.predict(te_0)
        # --- p(X) = Pr_s(D=1 | X): propensity score ---
        pi_m = clone(clf_model).fit(
            data_s.iloc[tr_s][X_cols], data_s.iloc[tr_s]["D"])
        pi_hat[te_s] = pi_m.predict_proba(
            data_s.iloc[te_s][X_cols])[:, 1]
        # --- s(S,X): density ratio via classification ---
        idx_all = np.r_[tr_l, tr_s + n_l]
        s_m = clone(clf_model).fit(
            data_all.iloc[idx_all][SX],
            data_all.iloc[idx_all]["E"])
        s_p = s_m.predict_proba(data_l.iloc[te_l][SX])
        s_ratio = (s_p[:, 1] / s_p[:, 0]) * (n_l / n_s)
        # --- pi(S,X) = Pr_s(D=1 | S, X): extended propensity ---
        a_m = clone(clf_model).fit(
            data_s.iloc[tr_s][SX], data_s.iloc[tr_s]["D"])
        a_p = a_m.predict_proba(data_l.iloc[te_l][SX])
        pi_l = pi_m.predict_proba(
            data_l.iloc[te_l][X_cols])[:, 1]
        # --- w_d(S,X) weights for long-term correction ---
        w1_hat[te_l] = s_ratio * a_p[:, 1] / pi_l
        w0_hat[te_l] = s_ratio * a_p[:, 0] / (1 - pi_l)
```

DR Surrogate ATE Estimator: Pseudocode — Estimation & Inference

```
# (continued from nuisance estimation)
# Compute the three terms of the DR-ATE formula:
# tau = E_s[g(1,X)-g(0,X)]
#     + E_s[ipw * (h(S,X) - g(D,X))]
#     + E_l[(w1 - w0) * (Y - h(S,X))]
D_s = data_s["D"].values
Y_l = data_l["Y"].values
# Simplified IPW for ATE:
#   1{D=1}/p(X) - 1{D=0}/(1-p(X)) = (D-p(X))/(p(X)(1-p(X)))
ipw = (D_s - pi_hat) / (pi_hat * (1 - pi_hat))
term1 = np.mean(g1_hat - g0_hat)             # plug-in
term2 = np.mean(ipw * (h_hat_s - g_hat))      # g-debiasing
term3 = np.mean(
    (w1_hat - w0_hat) * (Y_l - h_hat_l))      # h-debiasing
tau = term1 + term2 + term3
# Influence function for ATE (two-sample)
psi_s = (g1_hat - g0_hat) \
      + ipw * (h_hat_s - g_hat) - tau          # n_s-vector
psi_l = (w1_hat - w0_hat) * (Y_l - h_hat_l)   # n_l-vector
# Asymptotic variance: Var(tau) = Var(psi_s)/n_s + Var(psi_l)/n_l
se = np.sqrt(np.var(psi_s) / n_s
           + np.var(psi_l) / n_l)
return tau, se, tau - 1.96 * se, tau + 1.96 * se
```

Several features of this implementation deserve emphasis:

1. **Coupled cross-fitting.** The `KFold` object is created once over the combined dataset and reused for all nuisance models. This ensures that the same fold partition is used for all models, which is essential for the validity of the cross-fitting procedure.

2. **Density ratio via classification.** The surrogate score $s(S, X)$ is estimated by training a classifier to distinguish between samples from the long-term and short-term datasets. The predicted probabilities are then converted to a density ratio using the formula $s(S, X) = \frac{\Pr(E=s|S,X)}{\Pr(E=\ell|S,X)} \cdot \frac{n_\ell}{n_s}$.

3. **Two-sample influence function.** The standard error is computed using the influence function, which has two components: one from the short-term data (ψ_s) and one from the long-term data (ψ_l). The asymptotic variance is $\text{Var}(\hat{\tau}) = \text{Var}(\psi_s)/n_s + \text{Var}(\psi_l)/n_l$, reflecting the fact that the estimator uses information from both datasets.

4. **Flexible model choice.** The implementation accepts arbitrary regression and classification models (`reg_model` and `clf_model`), allowing the use of linear models (e.g., Ridge, Logistic Regression) or nonlinear models (e.g., Gradient Boosting) depending on the complexity of the data-generating process.

5 Summary and Key Takeaways

This chapter introduced surrogate variable analysis, a powerful and practical framework for estimating long-term treatment effects from short-term data. By leveraging historical data to model the relationship between short-term surrogates and long-term outcomes, we can make timely and informed decisions without waiting for the full effects of an intervention to materialize.

1. **The Surrogacy Principle.** The core idea is to find short-term variables (S) that completely mediate the effect of a treatment (D) on a long-term outcome (Y). This allows us to identify the long-term effect by estimating the effect on a *forecast* of the outcome, learned from historical data. The surrogate index model $h(S, X) = \mathbb{E}_\ell[Y \mid S, X]$ is the key object that bridges the short-term and long-term data.

2. **Two Datasets, One Key Assumption.** Surrogate analysis requires two datasets: a short-term experimental or observational study (where D and S are observed), and a long-term historical study (where S and Y are observed). The validity of the approach hinges on the **invariance** assumption: the relationship between surrogates and the long-term outcome must be stable across the two datasets. Practical validation strategies include using historical experiments with growing windows of surrogate data.

3. **Beware Confounded Surrogates.** The identification strategy can fail if the surrogates themselves are confounded by unobserved variables that also affect the long-term outcome. Conditioning on a confounded surrogate opens a collider path, invalidating the key conditional independence. In such cases, the analysis must also control for these confounders, or more advanced techniques (e.g., using other experiments as instruments, or proximal causal inference methods) must be used to isolate the causal relationship between the surrogates and the outcome.

4. **Doubly Robust Estimation.** The nested structure of the surrogate estimand lends itself to a doubly robust estimation strategy with recursive debiasing. The first correction term debiases for the outer regression g using standard IPW. The second correction term debiases for the surrogate index h using a combination of the density ratio (surrogate score) and a projected IPW, and is computed on the long-term data. The resulting estimator is consistent if either the outcome models or the propensity/density models are correctly specified.

5. **Practical Implementation.** The estimator is implemented with coupled cross-fitting over the combined short-term and long-term datasets. Standard errors are computed using a two-sample influence function. The Monte Carlo simulation confirms that the estimator is unbiased, has well-calibrated standard errors, and achieves nominal coverage.

Surrogate analysis is an indispensable tool for any data scientist or researcher working on problems with long-term outcomes. It bridges the gap between the need for timely decisions and the slow-moving nature of the phenomena we wish to understand, enabling a more agile and data-informed approach to causal inference.

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